HH-I-524C
October 1, 1980
SUPERSEDING
HH-I-524B
November 6, 1972

FEDERAL SPECIFICATION

INSULATION BOARD, THERMAL (POLYSTYRENE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use by all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the requirements for expanded polystyrene, thermal insulation boards. The thermal insulation boards, in this specification, have a service temperature range from -65° Farenheit (F) to +165° F. This specification does not cover cryogenic applications. For intended use see 6.1.

1.2 Classification. The thermal insulation boards covered by this specification shall be of the following types, as specified in 6.2, and shall meet the requirements of 3.3, table I.

Type I		
	Compression strength	10.0 pounds per square inch (psi) minimum.
	Density	0.9 pounds per cubic foot (pcf) minimum.
	Water absorption	4.0 (percent volume) maximum.
Type II		
	Compression strength	15.0 psi minimum.
	Density	1.3 pcf minimum.
	Water absorption	3.0 (percent volume) maximum.
Type III		
	Compression strength	20.0 psi minimum.
	Density	1.6 pcf minimum.
	Water absorption	2.0 (percent volume) maximum.
Type IV		
	Compression strength	20.0 psi minimum.
	Density	1.6 pcf minimum.
	Water absorption	0.3 (percent volume) maximum.
Type V		
	Compression strength	100.0 psi minimum.
	Density	3.0 pcf minimum.
	Water absorption	0.3 (percent volume) maximum.

FSC 5640

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2. APPLICABLE DOCUMENTS

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston; New York; Washington, DC; Philadelphia; Atlanta; Chicago; Kansas City, MO; Fort Worth; Houston; Denver; San Francisco; Los Angeles; and Seattle, WA.

(Federal Government activities may obtain copies of Federal specifications, standards, and commercial item descriptions, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

2.1 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

C177 - Standard Test Method for Steady-State Thermal Transmission Properties by Means of the Guarded Hot Plate.
C203 - Standard Test Method for Breaking Load and Calculated Flexural Strength of Preformed Block-Type Thermal Insulation.
C272 - Standard Test Method for Water Absorption of Core Material for Structural Sandwich Constructions.
C303 - Standard Test Method for Density of Preformed Block-Type Thermal Insulation.
C355 - Standard Test Method for Water Vapor Transmission of Thick Materials.
C390 - Sampling Preformed Thermal Insulation.
C518 - Standard Test Method for Steady-State Thermal Transmission Properties by Means of the Heat Flow Meter.
C550 - Standard Test Methods for Trueness and Squareness of Block Thermal Insulation.
D1621 - Standard Test Method for Compressive Properties of Rigid Cellular Plastics.
D2126 - Standard Test for Response of Rigid Cellular Plastics to Thermal and Humid Aging.
D2863 - Measuring the Minimum Oxygen Concentration to Support Candlelike Combustion of Plastic (Oxygen Index).

E96 replaced in 82'

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19102.)

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National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Materials. Cellular expanded polystyrene thermal insulation boards shall be formed by the expansion of polystyrene resin bead or granules in a closed mold or by the expansion of polystyrene base resin in an extrusion process. The thermal insulation boards shall be of uniform density, homogeneous, and essentially unicellular. All thermal insulation boards shall contain sufficient flame retardance to meet the oxygen index requirements of 3.3, table I, (see 6.4).

3.2 First article. When specified (see 6.2), the contractor shall furnish a complete thermal insulation board of the type, and size specified for first article inspection and approval (see 4.2.1 and 6.7).

3.3 Physical properties. The thermal insulation boards, when conditioned and tested in accordance with 4.5 through 4.5.11, shall meet the physical properties specified in table I.

TABLE I. Physical properties.

Property	Type I	Type II	Type III	Type IV	Type V	Test Reqmt. Para.
Compression strength (psi) minimum	10.0	15.0	20.0	20.0	100.0	4.5.6
Density (pcf) minimum	0.9	1.3	1.6	1.6	3.0	4.5.3
Dimensional stability (percent change) maximum	2.0	2.0	2.0	2.0	2.0	4.5.4
Oxygen index (percent) minimum	24.0	24.0	24.0	24.0	24.0	4.5.7

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TABLE I. Physical properties.

Property	Type I	Type II	Type III	Type IV	Type V	Test Reqmt. Para.
Water vapor ~~transmission~~ PERMEANCE (perm-inch) maximum	5.0	3.5	1.5	1.1	1.1	4.5.5
Water absorption (percent volume) maximum	4.0	3.0	2.0	0.3	0.3	4.5.8
Flexural strength (psi) minimum	25.0	40.0	55.0	60.0	100.0	4.5.9
*Thermal resistance Minimum at 75° ±2 F Mean Temperature	3.6	4.0	4.4	5.0	5.0	4.5.10
*Based on 1-inch board thickness						

3.3.1 Thermal resistance (R-value). The actual R-value when tested in accordance with 4.5.10 shall not vary more than 10 percent from the labeled value. The R-value shall be marked on the shipping container or board as required in 5.2 and 5.2.1.

3.3.2 Bead fusion. Molded foam boards shall have an essentially fused bead structure. When tested in accordance with 4.5.11, more than one-half of the exposed surface area shall have sheared or broken beads indicating that those beads have been fused.

3.4 Dimensions. The thermal insulation boards shall meet the tolorance and dimensional requirements of 3.4.1 through 3.4.4 when examined in accordance with 4.4 and 4.4.1.

3.4.1 Dimensional tolerances. Unless otherwise specified (see 6.2), length tolerance shall not exceed ± 1/32 inch per foot of length, width tolerance shall not exceed ± 1/16 inch per foot of width, and the thickness shall not exceed a tolerance of ± 1/16 inch per inch of thickness.

3.4.2 Length. The length of the thermal insulation boards shall be as specified (see 6.2 and 6.6).

3.4.3 Width. The width of the thermal insulation boards shall be as specified (see 6.2 and 6.6).

3.4.4 Thickness. The thickness of the thermal insulation boards shall be as specified (see 6.2, 6.5 and 6.6).

3.5 Edges and corners. The thermal insulation boards shall meet the requirements of 3.5.1 through 3.5.3 when examined in accordance with 4.4 and 4.4.1.

3.5.1 Board squareness. The thermal insulation boards shall not be out of square more than 1/16 inch per foot of width or length.

3.5.2 Straight edges. Unless otherwise specified (see 6.2), the thermal insulation boards shall be furnished with straight edges. Edges shall not deviate more than 1/32 inch per foot.

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3.5.3 Shiplap, and tongue and groove edges. When specified (see 6.2), the thermal insulation boards shall be furnished with either shiplap or tongue and groove edges. For insulation boards manufactured with shiplap edges, the depth of the shiplap edges shall be one-half the thickness of the board. For boards less than 2 inches of thickness the width of the lap shall be one-half the thickness of the board. For boards equal to or greater than 2 inches of thickness the width of the lap shall be 1 inch. The resulting joint shall be smooth and uniform. The tongue of one board shall fit snugly into the groove of a second board. The resulting joint shall be smooth and uniform.

3.6 Flatness. The thermal insulation boards shall not depart from absolute flatness more than 1/32 inch per foot of length or width when examined in accordance with 4.4 and 4.4.1.

3.7 Marking. The individual thermal insulation boards shall be marked with the following information: type, this specification number (HH-I-524C), and the manufacturer's name or trademark. An alternate method to that of marking the individual boards shall be the color striping of the individual boards. Marking shall be examined in accordance with 4.4.

3.7.1 Alternate method. When the alternate method of color striping is used, the stripes shall be a permanent color and shall not degrade the board's physical properties. Stripes shall be applied to one end and one side of the board, across the thickness, and shall be near the left-hand corner as viewed when facing the board's striped end or side.

3.7.1.1 First stripe. The first stripe shall denote this specification number (HH-I-524C). The stripe shall be a minimum of 1/2-inch wide and shall be violet in color.

3.7.1.2 Second stripe. The second stripe shall be located to the right of the first stripe and separated sufficiently to prevent intermix of colors. The stripe shall be a minimum of 1/2-inch wide and denote the basic insulation combinations as following:

Type I - Black
Type II - Magenta
Type III - Yellow
Type IV - Brown
Type V - Orange

3.8 Standard commercial product. The thermal insulation boards of the same classification shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the thermal insulation boards being furnished. A standard commercial product is a product which has been sold or is currently being offered for sale, on the commercial market through advertisements or manufacturer's catalogs or brochures.

3.9 Workmanship. The thermal insulation boards shall have no defects which will adversely affect their service qualities. The boards shall be of uniform texture, free from the accumulation of unexpanded materials, foreign inclusion, broken edges and corners, slits, and objectional odors when examined in accordance with 4.4.

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3.9.1 Crushing and depressions. The thermal insulation board shall have no crushed or depressed areas on any surface exceeding 1/8-inch in depth on more than 10 percent of the total surface area when examined in accordance with 4.4.

3.9.2 Voids. The thermal insulation boards shall have no more than eight voids, having dimensions larger than 1/8-inch by 1/8-inch by 1/8-inch, per 8 square feet of surface area of the board when examined in accordance with 4.4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).
(c) Preparation for delivery inspection (see 4.8).

4.2.1 First article inspection. First article inspection shall be performed on one insulation board when a first article sample is required (see 6.2 and 6.7). This inspection shall include the examination of 4.4 and the tests of 4.5. The first article may be a first production item or a standard production item from the contractor's current inventory provided the insulation board meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining thermal insulation boards to be furnished under the contract.

4.2.1.1 First article tests. When first article is required (see 6.2), the first article shall be tested as specified in 4.5 through 4.5.11. The first article tests shall be performed by the contractor under the direction and in the presence of Government representatives. Failure to pass any phase of the required tests shall be cause for rejection.

4.2.2 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.4 and 4.4.1 and the tests of 4.5 through 4.5.11.

4.2.2.1 Inspection lot. Unless otherwise specified (see 6.2), the inspection lot and lot size shall be in accordance with ASTM method C390.

4.2.2.2 Sampling for visual and dimensional examination. Unless otherwise specified (see 6.2), insulation boards shall be randomly selected from each lot offered for delivery to the Government in accordance with ASTM method C390.

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4.2.2.3 Sampling for tests. Unless otherwise specified (see 6.2), the thermal insulation boards covered by this specification shall be sampled in accordance with ASTM method C390.

4.3 Unit of product. The unit of product for visual examination and tests shall be one insulation board.

4.4 Examination. Each thermal insulation board shall be examined for compliance with the requirements specified in section 3 of this specification. Any redesign or modification of the contractor's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection. The examination shall be conducted as specified in table II and as otherwise specified in this document.

TABLE II. Classification of defects.

Categories	Defects	Reference paragraph
Major:		
101	Type and class not as specified	1.2
102	Dimensions - length, width, thickness not as specified or not within tolerance	3.4
103	Edges and corners - not as specified or within tolerance	3.5
104	Flatness - not within tolerance	3.6
105	Marking - not as specified	3.7, 5.2 and 5.2.1
106	Foreign inclusions	3.9
107	Broken edges or corners, slits, objectional odors, and broken boards	3.9
108	Unexpanded materials	3.9
109	Crushed or depressed areas exceeding 1/8-inch in depth on more than 10 percent of the total surface area of the board	3.9.1
110	More than eight voids per 8 square feet of surface area having dimensions larger in size than 1/8 inch by 1/8 inch by 1/8-inch in depth.	3.9.2
Minor:		
111	Crushed and depressed areas exceeding 1/8-inch in depth on not more than 10 percent of the total surface area of the board	3.9.1

4.4.1 Dimensions, trueness, squareness, and flatness. Sample boards selected in accordance with 4.2.2.2 shall have the length, width, and thickness dimensions measured in accordance with ASTM C303, and shall meet the requirements of 3.4.1 through 3.4.4. The trueness, squareness and flatness shall be determined in accordance with ASTM C550 except that a straight edge of adequate length shall be used. The boards shall meet the requirements of 3.5.1 and 3.6.

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4.5 Tests. Thermal insulation boards, after conditioning and aging in accordance with 4.5.1 and 4.5.2, shall be tested as specified in 4.5.3 through 4.5.11. In the event of conflict between ASTM test procedures and conditions, and test procedures and conditions as stated in this document, the test procedures and conditions in this specification shall govern. Tests for compressive strength, water absorption, and flexural strength shall be performed on test sample boards that are representative of the type of boards ordered, except, the thickness shall be 1 inch. Boards that are manufactured and sold with skins shall be tested with skins intact. When boards are tested with "skins-in-place", this condition will be noted in the report.

4.5.1 Conditioning. Sample boards to be used for the density test, dimensional stability test, water vapor transmission test, and bead fusion test shall be conditioned at 73° F ± 4° F and 50 percent ± 5 percent relative humidity for a minimum of 40 hours prior to the start of tests of 4.5.3 through 4.5.5 and 4.5.11. Sample boards to be used for the compressive strength test, oxygen index test, water absorption test, flexural strength test, and thermal conductance test, shall be conditioned as specified in the applicable test procedure.

4.5.2 Aging. Thermal insulation boards that incorporate a blowing agent other than air or pentane, shall be aged for either 90 days at 140° F ±2° F dry heat or 6 months at ambient condition prior to conditioning and thermal resistance testing of 4.5.10. Aging shall be performed on test sample boards that are representative of the type of boards ordered, except the thickness shall be 1inch. Boards that are manufactured and sold with skins shall be aged with skins intact. When boards are aged with "skins-in-place", this condition will be noted in the report. Air circulation shall be provided so that all surfaces of the boards are exposed to the surrounding environmental conditions.

4.5.3 Density test. The thermal insulation boards shall be tested in accordance with ASTM C303 and shall meet the requirements of 3.3, table I.

4.5.4 Dimensional stability test. The thermal insulation boards shall be tested for dimensional stability in accordance with ASTM D2126 at 158° F ± 4°F, 97 percent ± 3 percent relative humidity, and -40° F ± 2° F, ambient relative humidity. The test period shall be 7 days. The thermal insulation boards shall meet the requirements of 3.3, table I.

4.5.5 Water vapor ~~transmission~~ permeance test. The thermal insulation boards shall be tested in accordance with ASTM C355, dessicant method, and shall meet the requirements of 3.3, table I.

4.5.6 Compressive strength test. The thermal insulation boards shall be conditioned and tested in accordance with procedure A of ASTM D1621 at 10 percent deformation or yield, which ever occurs first. The compressive strength shall be determined on a 3-inch by 3-inch by 1-inch specimen. The compressive strength shall meet the requirements of 3.3, table I.

4.5.7 Oxygen index test. The thermal insulation boards shall be conditioned and tested in accordance with ASTM D2863, and shall meet the requirements of 3.3, table I. The testing laboratory temperature shall be 73° F ± 4° F.

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4.5.8 Water absorption test. The thermal insulation boards shall be conditioned and tested in accordance with ASTM C272, and shall meet the requirements of 3.3, table I. The soak time shall be 24 hours, and the test specimen size shall be 12 inches by 12 inches by 1 inch thick.

4.5.9 Flexural strength test. The thermal insulation boards shall be conditioned and tested in accordance with ASTM C203, and shall meet the requirements of 3.3, table I. The specimen shall be 12 inches long by 3 inches wide by 1 inch thick.

4.5.10 Thermal conductance/resistance test. The thermal insulation boards after aging in accordance with 4.5.2 shall be conditioned in accordance with ASTM C518 and tested in accordance with either ASTM C177 or C518, and shall meet the requirements of 3.3, table I and 3.3.1. The mean testing temperatures shall be 75° F ± 2° F and with a differential temperature of 50° F ± 2° F.

4.5.11 Bead fusion test (molded foam boards only). The bead fusion test shall be performed on a 6-inch by 6-inch test specimen that is cut from the thermal insulation board. The test specimen shall be broken into two approximately equal parts by scoring around the specimen to a scoreline depth not exceeding 10 percent of the thickness, and then breaking the specimen by hand pressure. The two broken surfaces (not including the scored areas) shall then be examined to determine whether the area of broken and sheared beads is greater than that of the unbroken, discrete beads (see 3.3.2).

4.6 Certificate of compliance. Unless otherwise specified (see 6.2), a certificate of compliance will be accepted for individual lot shipments in lieu of the completion of the individual test requirements of 4.5.3 through 4.5.11.

4.6.1 Criteria for acceptance. For a certificate of compliance to be accepted, first article requirements of 4.2.1 must have been met, and there shall have been no changes in the manufacturing techniques or materials that affect the physical properties since the examinations and tests were last performed.

4.7 Certificate of compliance on aged boards. A certificate of compliance will be accepted in lieu of aging as specified in 4.5.2

4.7.1 Criteria for acceptance. For a certificate of compliance on aged boards to be accepted, the boards must have been aged as specified in 4.5.2 within the last 2 years, and there shall have been no changes in the manufacturing techniques or materials that affect the physical properties during or since the aging was performed.

4.8 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing, and marking comply with the requirements of section 5 of this specification.

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5. PREPARATION FOR DELIVERY

5.1 Commercial. Unless otherwise specified (see 6.2, and 6.8) the thermal insulation boards shall be packaged in accordance with normal commercial practice. The complete pack shall be designed to protect the boards against damage during shipment, handling and storage, insure delivery at destination, provide for redistribution by the initial receiving activity, and be acceptable by common carrier under the National Motor Freight Classification rules or Uniform Freight Classification rules.

5.2 Marking. Unless otherwise specified (see 6.2), the marking of shipping containers shall be as specified in the contract or order.

5.2.1 Other marking. In addition to the marking required in 5.2, the following shall be marked on each shipping container or board:

(a) Insulation specification number
(b) Type
(c) Manufacturer's name or trademark
(d) Minimum certified R-value
(e) Instructions concerning the certified R-value at 75° F mean temperature for the thermal insulation thickness supplied, as following:

"R means the resistance to heat flow; the higher the value, the greater the insulation power. This insulation must be installed properly to get the marked R-value. Follow the manufacturer's instructions carefully. If a manufacturer's fact sheet is not provided with the material shipment, request it, and review it carefully."

(f) This note: "WARNING" - This product is combustible. Do not expose to any open flame or ignition sources. The product should be installed in accordance with manufacturer's instructions.

6. NOTES

6.1 Intended use. The polystyrene insulation boards covered by this specification are intended for use as thermal insulation primarily in building perimeters, ceilings, roof, walls, and other applications where the insulation will not be subjected to temperatures below -65° F or above 165° F (see 1.1). The ultimate efficiency of this thermal insulation depends to a large degree on how it is used. Therefore, it is important the manufacturer's instructions and design data be reviewed for each application.

6.1.1 Cryogenic applications This specification does not cover cryogenic applications. Consult manufacturer for recommendations and physical properties of polystyrene insulation boards in cryogenic conditions.

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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type (see 1.2).
(c) If first article is required for inspection and approval (see 3.2, 4.2.1, 4.2.1.1, and 6.7).
(d) R-value, if other than 1-inch board (see 3.3, table I and 6.5).
(e) Tolerance, if other than specified (see 3.4.1).
(f) Length and width required (see 3.4.2, 3.4.3, and 6.6).
(g) Thickness required (see 3.4.4, 6.5, and 6.6).
(h) If other than straight edges are required (see 3.5.2 and 3.5.3).
(i) If lot or lot size is different (see 4.2.2.1).
(j) Sampling, if different (see 4.2.2.2, and 4.2.2.3).
(k) If a certificate of compliance is not acceptable (see 4.6).
(l) If packing is other than specified (see 5.1, 6.8).
(m) If marking is other than specified (see 5.2, 6.8).

6.3 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provisions of paragraph 7-104.9(n) of the Defense Acquisition Regulation (DAR), the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data specified below shall be delivered in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data requirements	Applicable DD 1664
4.6 and 4.7	Certificate of compliance	DI-E-2121

6.4. Fire characteristics. Polystyrene is an organic material and is therefore combustible. It should not be exposed to open flames or other ignition sources. The fire performance of the material should be addressed through fire test requirements established by the appropriate governing documents, some of which are:

(a) Factory Mutual Research - FMR 4450 Approval Standard for Class I Insulation Steel Roofs.
(b) Factory Mutual Research - FMR 4880 Building Corner Fire Tests.
(c) ANSI/ASTM E119 Fire Tests of Building Construction and Materials
(d) ANSI/ASTM E84 Surface Burning Charecteristics of Building Materials.
(e) Uniform Building Code Standard No. 42-1 Test Method for Fire Hazard Classification of Building Materials.

The values obtained by the Oxygen Index Test (see 3.3, table I and 4.5.7) do not indicate or describe the fire hazard of the materials, and is used in this specification only to distinguish between insulation boards formulated with a flame retardant and those not so formulated.

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6.5 Thermal resistance - thickness relationship. The thermal resistance (R-value) of the thermal insulation boards supplied will vary with thickness. Therefore when ordering specify the R-value or the thickness, but not both. R-value versus thickness is not a straight line function. For additional information see ASTM C177 or C518.

6.6 Availability. The insulation boards are generally available in the sizes given in table III.

TABLE III. Standard sizes.

	Types I, II, and III	Type IV,V
Width (inches)	12 to 48	16, 24, 48
Length (inches)	48 to 192	48, 96, 108, 144
Thickness (inches)	3/8 to 24	3/4 to 3-1/2

6.7 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the DAR. The first article should be a first production item consisting of one complete thermal insulation board or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The contracting officer should include specific instructions in all procurement documents regarding arrangement for examinations, tests, and approval of the first article.

6.8 Military preparation for delivery requirements. The requirements in 5.1 are designed primarly for commercial type shipments. When additional requirements for level A or B military shipments are required, the necessary instructions will be included in the contract or order.

6.9 Cross reference.

HH-I-524B	HH-I-524C
Type I, class A	Type I
Type II, class A	Type II
None	Type III
Type II, class B	Type IV
Type III, class B	Type V

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MILITARY CUSTODIAN:

Navy - YD

Review activity:

Army - CE

User activities:

Army - ME
Navy - OS

CIVIL AGENCY COORDINATING ACTIVITIES:

COM - NBS
DOT - SMS
GSA - FSS, PCD
HUD - TCS

Preparing activity:

Navy - YD

Project No. 5640-0306

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